

CO4 – ASSESSMENT TASK 3

COMPARATIVE ANALYSIS

Quantitative Evaluation of Computer Vision Techniques

Object Detection • Motion Estimation • Depth Estimation

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Course Outcome: CO4 – Data Interpretation & Comparative Evaluation in Computer Vision

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1. Introduction and Objective

This report performs a quantitative comparative analysis of computer-vision techniques across three distinct problem areas: object detection, motion estimation, and depth estimation. In each case, two competing methods are evaluated using the standard numerical metrics of that sub-field — precision and recall for detection, average displacement error for motion tracking, and absolute error for depth estimation — and the results are used to logically justify which method performs better.

The objective is to demonstrate not only the correct calculation of each metric but also the reasoning process used to translate raw numbers into a defensible engineering decision: what a given precision or error value actually means for the reliability of a vision system in practice, and why one method is preferable to another once the numbers are interpreted rather than just computed.

2. Background: Evaluation Metrics Used in this Analysis

2.1 Precision and Recall (Object Detection)

Precision measures how many of the objects a model flagged as detections were actually correct, while recall measures how many of the true objects present in the scene the model actually managed to find. A model can achieve high precision by being overly cautious (missing real objects to avoid false alarms) or high recall by being overly liberal (flagging many false positives to avoid missing anything) — which is why both must be examined together, and why the F1-score, their harmonic mean, is often used to summarize overall detection quality in a single number.

$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN) \quad F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Here, TP (True Positives) are correctly detected objects, FP (False Positives) are false alarms, and FN (False Negatives) are real objects the model missed.

2.2 Average Displacement Error (Motion Estimation)

In optical-flow / motion-tracking systems, displacement error measures the distance in pixels between the position an algorithm predicted for a tracked point and its true position in the next frame. Averaging this error

across several consecutive frames gives a stable measure of how accurately a method tracks motion over time — lower average error means the estimated trajectory stays closer to the true motion path.

$$\text{Average Displacement Error} = (\Sigma \text{ errors across frames}) / (\text{number of frames})$$

2.3 Absolute Error (Depth Estimation)

For depth-sensing systems, absolute error is simply the magnitude of the difference between an estimated depth and the true (ground-truth) depth of an object, ignoring the sign of the difference. Averaging this across multiple objects gives the Mean Absolute Error (MAE), a standard way to summarize a depth sensor's overall accuracy.

$$\text{Absolute Error} = | \text{Estimated Depth} - \text{Actual Depth} | \quad \text{MAE} = (\Sigma \text{ absolute errors}) / (\text{number of objects})$$

3. Problem Statements (as given)

3.1 Object Detection Performance Comparison

Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives.

- (a) Calculate the precision and recall for both models.
- (b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

3.2 Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames.

- (a) Calculate the average displacement error for both methods.
- (b) Compare the two methods and determine which method provides more accurate motion estimation.
- (c) Justify your answer based on the error values.

3.3 Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects. The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively.

- (a) Calculate the absolute error for each object for both methods.
- (b) Calculate the average absolute error for both methods.
- (c) Compare the methods and identify which provides more accurate depth estimation.

4. Answer 1: Object Detection Performance Comparison

4.1 Given data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A (Viola-Jones)	80	20	30
Model B (YOLO)	90	15	20

4.2 (a) Precision and Recall calculation

Model A:

$$\text{Precision}(A) = 80 / (80 + 20) = 80 / 100 = 0.80 = 80\%$$

$$\text{Recall}(A) = 80 / (80 + 30) = 80 / 110 = 0.7273 \approx 72.73\%$$

Model B:

$$\text{Precision}(B) = 90 / (90 + 15) = 90 / 105 = 0.8571 \approx 85.71\%$$

$$\text{Recall}(B) = 90 / (90 + 20) = 90 / 110 = 0.8182 \approx 81.82\%$$

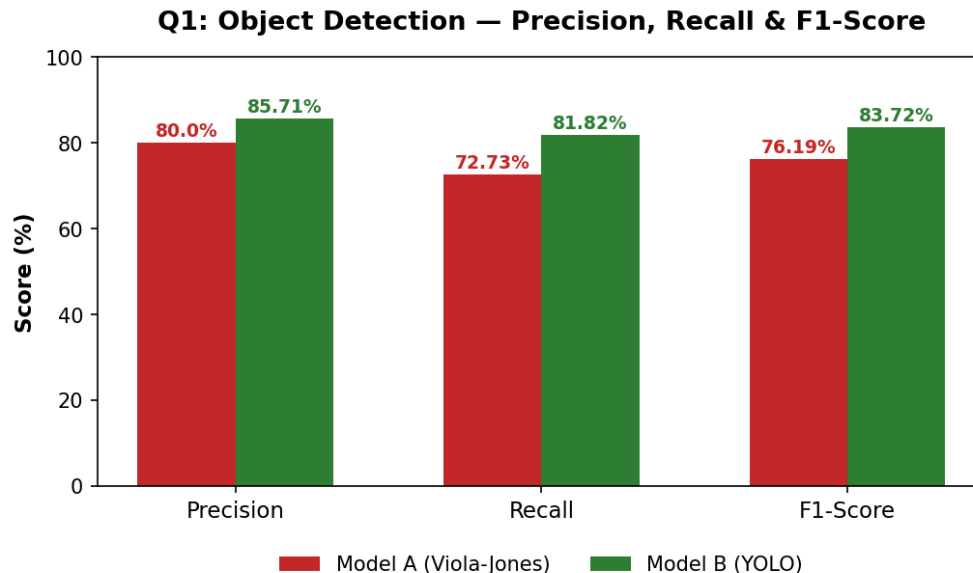
For a more complete picture, the F1-score (harmonic mean of precision and recall) is also computed for both models:

$$F1(A) = 2 \times (0.80 \times 0.7273) / (0.80 + 0.7273) = 0.7619 \approx 76.19\%$$

$$F1(B) = 2 \times (0.8571 \times 0.8182) / (0.8571 + 0.8182) = 0.8372 \approx 83.72\%$$

4.3 Summary table

Model	Precision	Recall	F1-Score
Model A (Viola-Jones)	80.00%	72.73%	76.19%
Model B (YOLO)	85.71%	81.82%	83.72%



4.4 (b) Comparison and Justification

Model B outperforms Model A on every single metric: its precision (85.71%) is higher, meaning a smaller fraction of its detections are false alarms; its recall (81.82%) is higher, meaning it misses fewer real objects; and consequently its F1-score (83.72%) — the balanced summary of both — is also higher than Model A's 76.19%. Because Model B is not winning on one metric at the expense of the other (which would require a trade-off judgement), but is strictly better on both precision and recall simultaneously, the comparison is unambiguous.

Model B (YOLO) provides better overall object detection performance. It detects more of the true objects present (higher recall, fewer missed detections) while simultaneously producing fewer false alarms relative to its total detections (higher precision). This combination — higher TP with proportionally lower FP and FN — reflects a genuinely more reliable detector rather than one that has simply shifted its errors from one type to another.

5. Answer 2: Motion Estimation Error Analysis

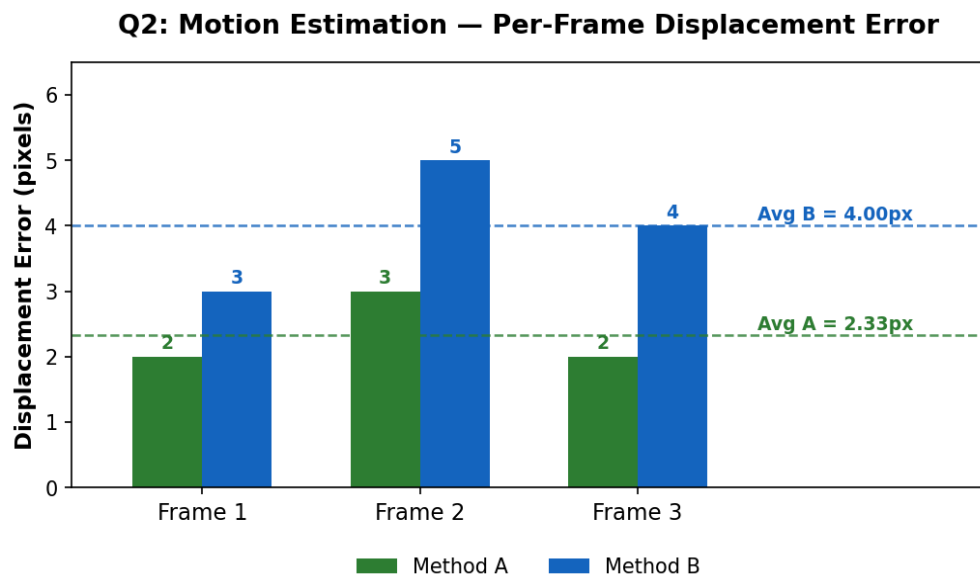
5.1 Given data

Method	Frame 1 Error (px)	Frame 2 Error (px)	Frame 3 Error (px)
Method A	2	3	2
Method B	3	5	4

5.2 (a) Average displacement error calculation

$$\text{Average Error}(A) = (2 + 3 + 2) / 3 = 7 / 3 = 2.33 \text{ px}$$

$$\text{Average Error}(B) = (3 + 5 + 4) / 3 = 12 / 3 = 4.00 \text{ px}$$



5.3 (b) Comparison

Method A has a lower average displacement error (2.33 px) than Method B (4.00 px) — a difference of 1.67 px, which represents roughly 42% lower error relative to Method B. Looking at the per-frame breakdown, Method A is also more consistent, with errors staying tightly between 2 and 3 pixels across all three frames, whereas Method B fluctuates more widely between 3 and 5 pixels, indicating less stable tracking from frame to frame.

5.4 (c) Justification

Method A provides more accurate motion estimation. Not only is its average error substantially lower, but its smaller frame-to-frame variance also suggests it is more numerically stable and therefore more trustworthy for applications where consistent tracking matters (e.g., feeding a Kalman filter or trajectory predictor downstream). Method B's higher and more variable error indicates it is more susceptible to noise or fast motion between frames, which would translate into a shakier, less reliable estimated trajectory in a real tracking pipeline.

6. Answer 3: Depth Estimation Comparison

6.1 Given data

Object	Actual Depth (m)	Stereo Vision Estimate (m)	SfM Estimate (m)
Object 1	2.0	2.0	2.5
Object 2	3.0	3.0	3.8
Object 3	4.0	4.0	5.0

6.2 (a) Absolute error per object

Object 1: $|2.0 - 2.0| = 0.0$ m (Stereo) $|2.5 - 2.0| = 0.5$ m (SfM)

Object 2: $|3.0 - 3.0| = 0.0$ m (Stereo) $|3.8 - 3.0| = 0.8$ m (SfM)

Object 3: $|4.0 - 4.0| = 0.0$ m (Stereo) $|5.0 - 4.0| = 1.0$ m (SfM)

6.3 (b) Average absolute error (Mean Absolute Error)

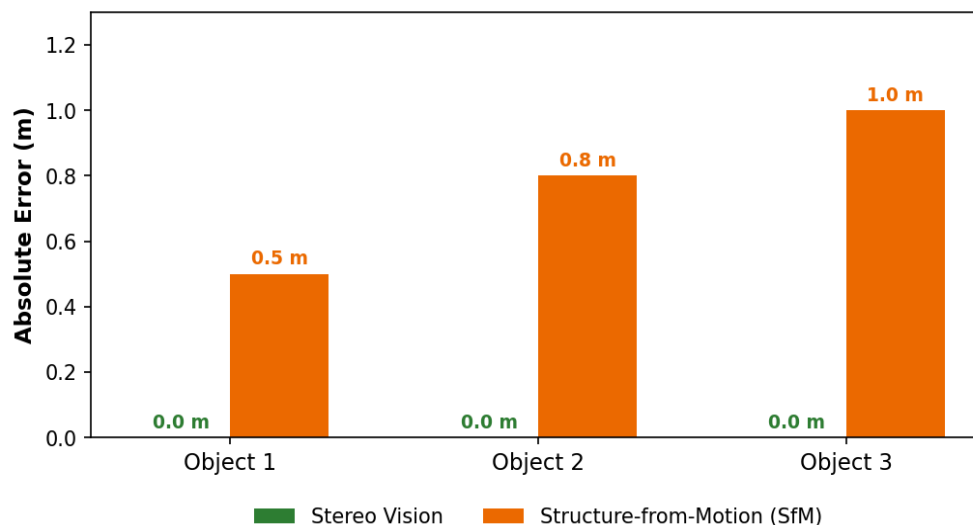
MAE (Stereo Vision) = $(0.0 + 0.0 + 0.0) / 3 = 0.0$ m

MAE (SfM) = $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.767$ m ≈ 0.77 m

6.4 Summary table

Method	Error – Object 1	Error – Object 2	Error – Object 3	MAE
Stereo Vision	0.0 m	0.0 m	0.0 m	0.0 m
SfM	0.5 m	0.8 m	1.0 m	0.77 m

Q3: Depth Estimation — Absolute Error per Object



6.5 (c) Comparison and Justification

Stereo Vision estimates match the actual depths exactly for all three objects (0.0 m error in every case), giving it a Mean Absolute Error of 0.0 m. Structure-from-Motion, in contrast, over-estimates depth for every object, with the error growing as depth increases (0.5 m at 2 m, 0.8 m at 3 m, and 1.0 m at 4 m), giving it a Mean Absolute

Error of approximately 0.77 m. This growing error pattern is a common characteristic of SfM systems, whose scale estimates can drift with distance in the absence of a fixed physical baseline, unlike stereo vision, which uses a known, fixed baseline between two cameras to triangulate depth directly and geometrically.

Stereo Vision provides more accurate depth estimation for this data. Its perfect agreement with ground truth (zero error on every object) versus SfM's substantial and worsening error makes this a clear and unambiguous result, not merely a marginal preference. In practice, this suggests that where two calibrated cameras are available (as in stereo vision), fixed-baseline triangulation should be preferred over single-camera SfM for tasks that demand precise metric depth, with SfM instead better suited to applications where only a single moving camera is available and relative (rather than exact metric) depth is acceptable.

7. Comparative Summary Across All Three Case Studies

Although these three case studies measure very different things — detection correctness, motion tracking precision, and depth accuracy — they share a common analytical structure: compute the relevant metric(s) for each method, compare the values directly, and let the numbers (not intuition) determine the winner. The table below consolidates the outcome of all three analyses.

Case Study	Metric(s) Used	Better Method	Key Evidence
1. Object Detection	Precision, Recall, F1	Model B (YOLO)	Higher on all three metrics (F1: 83.72% vs 76.19%)
2. Motion Estimation	Average Displacement Error	Method A	Lower & more stable error (2.33 px vs 4.00 px)
3. Depth Estimation	Absolute Error / MAE	Stereo Vision	Zero error vs 0.77 m MAE for SfM

A useful pattern across all three cases is that the better-performing method was better on every relevant metric simultaneously — there was no case here where one method won on one metric while losing on another, which is what makes each conclusion unambiguous rather than a judgement call between competing trade-offs. In practice, comparative analyses are not always this clean, so the same systematic approach (restate the data, compute every metric precisely, compare like-for-like, then justify) remains the correct method even when the winner is less clear-cut.

8. Conclusion

This comparative analysis applied standard, field-specific evaluation metrics to three computer-vision problems and reached a clear, numerically justified conclusion in each case: Model B (YOLO) is the stronger object detector, Method A is the more accurate and stable motion estimator, and Stereo Vision is the more accurate depth-estimation technique for the given data. Beyond the individual results, the exercise reinforces a broader principle of applied computer vision — that comparing techniques rigorously requires calculating the correct metric for the task at hand, interpreting what that metric means for real-world reliability, and basing the final decision on that evidence rather than on the raw sophistication or reputation of a method.

9. References

1. Powers, D. M. W. (2011). Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness & Correlation. *Journal of Machine Learning Technologies*.
2. Lucas, B. D., & Kanade, T. (1981). An Iterative Image Registration Technique with an Application to Stereo Vision. *Proceedings of Imaging Understanding Workshop*.
3. Hartley, R., & Zisserman, A. (2004). *Multiple View Geometry in Computer Vision*. Cambridge University Press.
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